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A Virtual-Reality Digital Twin Operator Display for Dynamic UAV Scene Reconstruction from In-Flight Detections

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Remote UAV operation requires the pilot to maintain spatial awareness of the aircraft, terrain, obstacles, and mission-relevant objects when direct visual contact or continuous high-quality video may be unavailable. This working manuscript presents Pipeline B, an implemented virtual-reality digital twin operator display in which a UAV pilot wearing a head-mounted display can inspect a dynamically reconstructed 3D representation of the operational scene. The system converts in-flight object detections into georeferenced semantic telemetry and uses those data to spawn, update, and remove dynamic actors inside an Unreal Engine and Cesium-based environment. The contribution is framed as a human-in-the-loop operator-display architecture, not as autonomous flight control, certified detect-and-avoid, or a replacement for regulatory safety channels. The paper currently defines the runtime contract, interaction concept, evidence map, and validation protocol required for a complete journal submission. Quantitative results are retained as explicit placeholders until the validation experiments, VR demonstration material, and hardware-dependent trials are completed.

Virtual reality; UAV; Digital twin; Scene reconstruction; Semantic telemetry; Teleoperation; Situational awareness

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1. Draft Status, Journal Target, and Placeholder Convention

This manuscript is a working architecture and evaluation draft for *Virtual Reality & Intelligent Hardware* (VRIH). The target article type is a full research paper rather than a rapid communication, because the argument requires architecture, runtime contract, prototype evidence, quantitative evaluation, and human-operator validation. VRIH rapid communications are capped at four printed pages, which is incompatible with the evidence burden of this paper.

The current version is not a submission-ready experimental article. It is a controlled draft that makes every missing result visible. Any unmeasured quantity is marked with [TBD: identifier and required evidence]. These placeholders must be replaced by measured results, figures, tables, or explicitly scoped limitations before journal submission.

The required placeholder identifiers are:

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- **TBD-DEMO**: screenshots, video captures, configuration records, and execution logs showing the implemented system operating in the VR display.
- **TBD-BW**: semantic telemetry bitrate, p95 bandwidth, packet rate, and comparison with H.264, H.265, or WebRTC video baselines.
- **TBD-LAT**: communication delay from the UAV or detection source to the pilot equipped with the VR headset, including Brain ingest, Unreal actor update, and visible headset refresh.
- **TBD-GEO**: two-dimensional and three-dimensional reconstruction error against ground truth.
- **TBD-LOSS**: behavior under controlled packet loss, jitter, and link interruption.
- **TBD-TRACK**: identity switches, stale-object duration, and track fragmentation.
- **TBD-HUMAN**: pilot task success, near misses, workload, and situational-awareness measures.
- **TBD-SAFETY**: operating limits, fallback rules, confidence thresholds, and uncertainty-display thresholds.
- **TBD-LOWVIS**: sensor-specific validation in night, smoke, fog, haze, glare, or other degraded-visibility conditions.

2. Motivation and Operator Task

Remote UAV pilots often depend on a video feed and a map view to infer the spatial relationship between the aircraft, terrain, obstacles, and mission-relevant objects. That interface can become insufficient when the video link is bandwidth-limited, delayed, compressed, interrupted, or constrained to a narrow camera field of view. The problem is not simply that video consumes bandwidth. The deeper problem is that the pilot must reconstruct a three-dimensional operational scene mentally from incomplete two-dimensional evidence.

Pipeline B addresses this human-interface problem by converting detected objects into a persistent, spatially organized digital-twin layer. The pilot-side display is intended for a head-mounted VR device, where the operator can inspect the reconstructed scene from an egocentric or external viewpoint. The task is not safety-critical flight control. The task is auxiliary situational awareness: keeping track of selected objects, obstacles, stale detections, and uncertainty while video, map, and semantic channels degrade differently.

The paper should therefore be evaluated around a concrete operator task:

> Given a UAV mission with terrain context, detected dynamic or static objects, and a constrained communication link, can a VR-equipped operator preserve better spatial awareness of mission-relevant entities than with video-only, map-only, or non-immersive hybrid displays?

The answer cannot be asserted from architecture alone. It requires demonstration evidence, communication measurements, geospatial validation, degradation tests, and human-operator assessment.

3. Related Work and Positioning

The contribution is not YOLO object detection, Unreal Engine simulation, Cesium geospatial rendering, ArduPilot telemetry, VR hardware, or UAV target geolocation in isolation. Those components are established. The contribution is the integration of in-flight detections, georeferenced semantic telemetry, runtime actor management, and immersive operator visualization into an evaluable VR digital-twin display.

Task-oriented and semantic communication already study transmitting task-relevant information rather than reconstructing full source signals, especially for edge video analytics [1]. Synthetic and hybrid vision systems for remotely piloted vehicles also predate this work; NASA's SmartCam3D combined live sensor information with synthetic visual context for improved operator awareness [2]. Digital-twin support for UAV teleoperation over constrained networks has been studied with field trials [3]. Unreal-based UAV simulation and MAVLink integration are

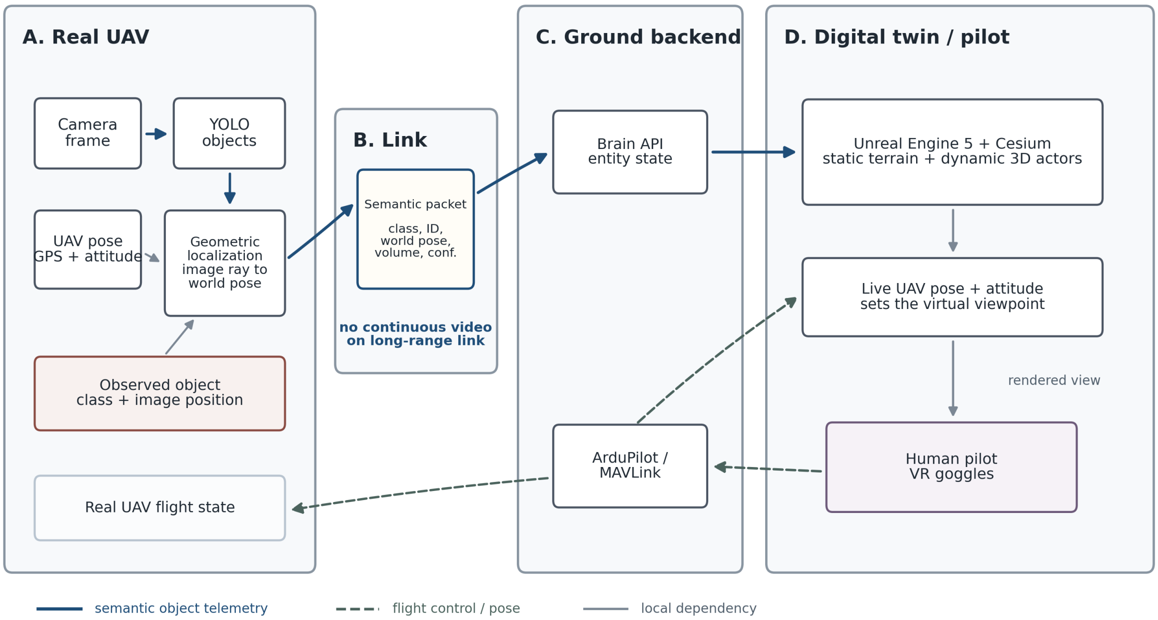

Figure 1 Pipeline B runtime flow. The UAV or upstream perception source detects objects with YOLO or another detector, estimates object world coordinates under calibrated camera and pose assumptions, and sends compact semantic telemetry to the ground station. Brain publishes the live entity layer, while Unreal Engine, Cesium terrain, ArduPilot telemetry, and the live object layer are fused into an auxiliary VR operator display. Solid arrows denote semantic object telemetry, dashed arrows denote flight-control and pose feedback, and thin gray arrows denote local dependencies.

established by systems such as AirSim [4]. UAV real-time object detection is a mature research area with extensive surveys [5]. Vision-based UAV target geolocation from camera pixels, pose, and attitude has also been studied with flight tests and filtering methods [6, 7]. BVLOS operations further require explicit treatment of detect-and-avoid, command-and-control reliability, operational envelopes, and regulatory compliance [8].

Pipeline B must therefore be evaluated as a human-in-the-loop VR operator-display architecture. The paper should not overclaim novelty over perception, networking, geolocation, or VR rendering individually. Its novelty claim must be narrower: an implemented runtime path and evaluation protocol for reconstructing UAV mission-relevant entities inside a VR digital twin from in-flight detections.

4. System Architecture

Pipeline B converts onboard or upstream object detections into a compact entity stream. For each relevant object, the system publishes an identity, class label, confidence score, estimated position, timestamp, and optional geometry. At the ground station, the Brain service normalizes and tracks these entities, then publishes the current entity layer to Unreal Engine. Unreal renders static terrain and map context through Cesium while spawning, updating, and despawning dynamic actors from the semantic entity stream.

The VR headset is the key operator-facing endpoint. The visual scene is not presented as an evidentiary truth layer. It is a synthetic operator display whose objects must carry confidence, age, source, and uncertainty semantics.

Figure 1 summarizes the runtime flow. The figure is an architecture diagram, not experimental evidence. Demonstration evidence must come from screenshots, videos, and logs from the real prototype; quantitative validation must come from communication-delay, bandwidth, geospatial-error, robustness, and user-task measurements.

4.1 Runtime Data Flow

The implemented runtime contract is organized around two HTTP interfaces:

1. A detection source publishes objects to `POST /api/obstacles`.
2. Brain normalizes source identity, class, confidence, timestamp, and position.
3. Brain exposes the current operator-display state through `GET /api/ui/data`.
4. `UPorceTelemetryComponent` in Unreal polls the state endpoint.
5. Unreal spawns, updates, or despawns actors using `entity_id` or `object_id`.
6. The VR-equipped operator observes the reconstructed entity layer in the digital twin.

Table 1 Pipeline B runtime data flow.

Stage	Producer	Consumer
Object detection	YOLO or detection source	Telemetry encoder
Estimated georeferencing	UAV pose + camera assumptions	Detection message
Object ingest	Detection sender	Brain endpoint
State publication	Brain	Unreal component
Actor reconstruction	Unreal component	VR operator display

4.2 Entity Contract

The essential telemetry fields are:

- stable `entity_id` or `object_id`;
- object class;
- confidence score;
- `lat`/`lon` geospatial position when available;
- `world_m` local coordinates when available;
- approximate footprint, volume, or class-specific geometry;
- timestamp, source identifier, and source metadata;
- stale-state and uncertainty fields for operator display [TBD: TBD-SAFETY freshness and uncertainty fields].

4.3 VR Operator Display Requirements

For VRIH, the display must be specified as an immersive operator interface rather than a generic 3D visualization. A publishable method section must define:

- headset or head-mounted display model and runtime [TBD: TBD-DEMO HMD configuration];
- viewpoint modes, such as UAV-centric, external chase, top-down inspection, or operator-selected viewpoints [TBD: TBD-DEMO viewpoint modes];
- visual encoding of class, confidence, age, and stale state [TBD: TBD-SAFETY display encoding];
- interaction model for inspecting entities, filtering classes, and reading uncertainty [TBD: TBD-HUMAN interaction model];
- fallback behavior when detections, pose, or network updates are stale [TBD: TBD-SAFETY fallback rule].

4.4 Georeferencing Assumptions

The current georeferencing claim must be treated as estimated georeferencing under calibrated camera and pose assumptions, not as a proven localization result. A publishable method section must specify:

- camera intrinsics, distortion model, and image resolution [TBD: TBD-GEO camera calibration];
- camera-to-body extrinsics and mount roll, pitch, and yaw [TBD: TBD-GEO extrinsic calibration];
- UAV pose, attitude, GNSS source, and timestamp synchronization [TBD: TBD-GEO synchronization error];
- terrain or ground-plane intersection assumptions [TBD: TBD-GEO terrain model];
- propagation of bounding-box, pose, attitude, and map uncertainty into object-position uncertainty [TBD: TBD-SAFETY uncertainty radius];
- operating envelope where projection error remains acceptable [TBD: TBD-SAFETY allowed operating envelope].

5. Current Readiness and Hardware Dependence

Some parts of the paper can be completed immediately from the existing software, logs, simulator, and controlled replay tools. These software-only tests are preparatory evidence; they are not sufficient on their own for a VRIH research-paper claim about a VR headset operator display. The submission-critical evidence still requires the real UAV or instrumented detection source, calibrated camera assumptions, VR headset, and human operators. Table 2 separates those two categories so that the remaining work is explicit.

Table 2 Work packages that can be completed now versus work requiring hardware.

Evidence item	Can be completed now	Requires hardware	Status
VRIH framing	Rewrite title, abstract, introduction, contribution, and conclusion around VR/HMD operator display.	No.	Done in this draft
Runtime contract	Document `POST /api/obstacles`, `GET /api/ui/data`, entity fields, actor lifecycle, and display semantics.	No.	Drafted
Bandwidth benchmark	Compute message size, update rate, mean bitrate, p95 bitrate, and packet rate from logs or synthetic/replayed detections.	Real mission logs improve validity but are not strictly required for first benchmark.	[TBD: TBD-BW]
Brain-to-Unreal latency	Instrument sender, Brain ingest, state publication, Unreal polling, and actor update on local or lab network.	HMD not needed for this partial latency segment.	[TBD: TBD-LAT partial]
Packet loss and jitter	Inject controlled delay, drop, and interruption into semantic telemetry and observe stale-state behavior.	No, unless validating over a real radio link.	[TBD: TBD-LOSS]
Synthetic tracking	Replay known trajectories and measure ID switches, stale duration, and fragmentation.	No for synthetic validation; yes for real motion validation.	[TBD: TBD-TRACK synthetic]
VR demonstration	Capture screenshots and video of pilot-side VR display running the implemented prototype.	Yes: HMD and running Unreal VR configuration.	[TBD: TBD-DEMO]
Drone-to-VR latency	Measure source event to visible headset update.	Yes: real or instrumented detection source plus HMD timing/capture.	[TBD: TBD-LAT e2e]
Geospatial error	Compare reconstructed object positions against surveyed or RTK/GNSS ground truth.	Yes: calibrated camera/pose and ground truth objects or controlled flight.	[TBD: TBD-GEO]
Human operator utility	Compare video-only, map-only, non-immersive hybrid, and VR digital twin conditions.	Yes: HMD, task scenarios, participants, and protocol approval if needed.	[TBD: TBD-HUMAN]
Low-visibility claims	Validate RGB/thermal/LiDAR/radar behavior in night, fog, glare, smoke, or equivalent conditions.	Yes: sensor and environment-specific trials.	[TBD: TBD-LOWVIS]

6. Claim-to-Evidence Map

Table 3 Claims retained in the paper and evidence required before submission.

Claim	Required evidence	Status	Acceptance criterion
A real VR digital-twin operator display has been developed.	Screenshots, videos, configuration records, and execution logs showing the UAV-to-Brain-to-Unreal/VR path operating end to end.	[TBD: TBD-DEMO]	Demonstration material shows object ingestion, actor update, and pilot-side VR visualization without relying only on mock diagrams.

Claim	Required evidence	Status	Acceptance criterion
Semantic telemetry can reduce communication load relative to video.	Bitrate comparison against fixed video baselines with identical mission duration.	[TBD: TBD-BW]	Semantic stream uses substantially less bandwidth while preserving required entity state.
Actor updates can be shown with low drone-to-pilot delay.	Synchronized timing from UAV or detector publication to Brain ingest, state publication, Unreal update, and VR-headset-visible actor refresh.	[TBD: TBD-LAT]	P95 latency stays below the operator-display threshold selected in the experiment.
Estimated object positions are operationally useful.	Ground-truth comparison over range, altitude, class, and terrain conditions.	[TBD: TBD-GEO]	Error distribution remains inside the declared uncertainty envelope.
The display degrades gracefully under packet loss.	Controlled loss, jitter, and interruption experiments against video-only and hybrid baselines.	[TBD: TBD-LOSS]	Stale entities are visible as stale and do not appear as fresh truth.
Entity persistence improves operator context.	Tracking metrics, stale-object duration, identity switches, and fragmentation.	[TBD: TBD-TRACK]	Identity stability improves without hiding missed detections.
The display helps human operators.	Controlled pilot study with task success, near misses, workload, NASA-TLX, SAGAT, SART, or equivalent awareness measures.	[TBD: TBD-HUMAN]	Operator benefit is statistically or operationally meaningful without increased overtrust.
Low-visibility rendering is useful only when sensors still detect hazards.	Sensor-specific trials for RGB, thermal, LiDAR, radar, or other inputs.	[TBD: TBD-LOWVIS]	Display explicitly distinguishes observed, inferred, stale, and missing objects.

7. Evaluation Protocol

7.1 Real-System Demonstration Artifacts

The first submission-critical evidence is not a metric but proof that the implemented prototype actually runs as claimed. The demonstration package should show the runtime chain from detections to Brain state to Unreal actor reconstruction to pilot-side VR display.

Table 4 Real-system demonstration placeholders.

Artifact	Purpose	Status	Acceptance criterion
Screenshots from the VR digital twin	Show the HMD-facing view, terrain context, reconstructed dynamic entities, confidence, and stale-state encoding.	[TBD: TBD-DEMO screenshots]	Captures correspond to the implemented system rather than illustrative mockups.
Video of system operation	Show live spawn, update, persistence, and despawn behavior during operation.	[TBD: TBD-DEMO video]	Video includes enough context to connect detections, telemetry, and VR visualization.
Runtime logs and configuration	Support reproducibility of the demonstrated setup.	[TBD: TBD-DEMO logs]	Logs identify endpoints, polling rate, timestamps, HMD runtime, headset mode, and display configuration.

7.2 Bandwidth Versus Video

This experiment can be prepared immediately from synthetic or replayed detection streams. It becomes stronger when repeated with real mission logs and synchronized video baselines.

Table 5 Bandwidth results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
Mean bitrate	H.264, H.265, WebRTC FPV	[TBD: TBD-BW mean bitrate]	TBD	Lower bitrate for same scenario duration
P95 bitrate	Same codec and link profile	[TBD: TBD-BW p95 bitrate]	TBD	Reduced p95 load without entity loss
Packet rate	Same mission duration	[TBD: TBD-BW packet rate]	TBD	Compatible with target link budget
Entity completeness	Ground-truth or replayed entity stream	[TBD: TBD-BW completeness]	TBD	Bandwidth reduction does not hide required entity state

7.3 End-to-End Latency

Latency must be decomposed because some segments can be measured now and some require the HMD. Brain ingest, state publication, and Unreal actor update can be instrumented without a real drone. Drone-to-pilot visible latency requires the HMD and a real or instrumented detection source.

Table 6 Latency results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
Detector publication to Brain ingest	Synchronized detector timestamp and Brain receive log	[TBD: TBD-LAT detector-to-brain]	TBD	Below selected runtime threshold
Brain to Unreal actor update	Polling and render logs	[TBD: TBD-LAT actor update]	TBD	P95 below operator-display threshold
Unreal update to VR headset display	Unreal timing, headset capture, or instrumented presentation timestamp	[TBD: TBD-LAT vr-display]	TBD	Visible update remains within the declared pilot-display threshold
End-to-end source-to-HMD visible update	Detector event to actor visible to the VR-equipped pilot	[TBD: TBD-LAT e2e-source-to-vr]	TBD	No unnoticed stale display

7.4 Geospatial Error

This validation is hardware-dependent if the paper claims real-world positioning accuracy. Synthetic coordinates or manually injected positions can test software consistency, but not camera-to-world georeferencing.

Table 7 Geospatial error results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
2D position error	RTK/GNSS or surveyed ground truth	[TBD: TBD-GEO 2D error]	TBD	Error inside declared uncertainty radius
3D position error	Ground-truth height or volume	[TBD: TBD-GEO 3D error]	TBD	Height error acceptable for visualization
Error by range	Near, mid, far detections	[TBD: TBD-GEO by range]	TBD	Error envelope reported by range bin
Uncertainty calibration	Measured error versus displayed uncertainty	[TBD: TBD-GEO uncertainty calibration]	TBD	Displayed uncertainty is not overconfident

7.5 Packet Loss and Jitter Robustness

This experiment can be implemented now with a network impairment layer or replay tool. It should test the semantic display against video-only and hybrid baselines, with special attention to stale-state visualization.

Table 8 Packet-loss and jitter results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
Packet loss degradation	Video-only and hybrid display	[TBD: TBD-LOSS loss sweep]	TBD	Graceful stale-state behavior
Jitter sensitivity	Fixed jitter profiles	[TBD: TBD-LOSS jitter sweep]	TBD	No false freshness under delayed updates
Interruption recovery	Link interruption scenario	[TBD: TBD-LOSS recovery]	TBD	Recovery state visible to operator
False freshness rate	Delayed or dropped updates	[TBD: TBD-LOSS false freshness]	TBD	Stale data is not presented as fresh state

7.6 Dynamic Object Tracking

Tracking can first be evaluated with replayed or synthetic trajectories. A stronger validation requires real moving objects, synchronized ground truth, and class-specific occlusion or detection-loss scenarios.

Table 9 Tracking results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
ID switches	Ground-truth trajectories	[TBD: TBD-TRACK ID switches]	TBD	ID switches reported and bounded
Track fragmentation	Moving-object trial	[TBD: TBD-TRACK fragmentation]	TBD	Fragmentation does not hide uncertainty
Stale-object duration	Controlled occlusion/loss	[TBD: TBD-TRACK stale duration]	TBD	Stale state visibly encoded
Despawn correctness	Known disappearance events	[TBD: TBD-TRACK despawn]	TBD	Actors are removed or marked stale according to declared policy

7.7 Human Operator Utility

For a VR-focused journal, the human-operator evaluation is not decorative. It is the evidence that the immersive display changes the operator task. A minimum study should compare video-only, map-only, non-immersive hybrid, and VR digital-twin conditions on the same mission scenarios.

Table 10 Human-operator study placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
Task success	Video-only, map-only, hybrid	[TBD: TBD-HUMAN task success]	TBD	Improvement or non-inferiority with lower link load
Spatial awareness	SAGAT, SART, or equivalent	[TBD: TBD-HUMAN awareness]	TBD	Operators better identify object position, age, and relevance
Workload	NASA-TLX or equivalent	[TBD: TBD-HUMAN workload]	TBD	Benefit without unacceptable cognitive load
Near misses or unsafe decisions	Same operator tasks	[TBD: TBD-HUMAN near misses]	TBD	No increase in safety-relevant events
Overtrust	Post-task questionnaire and stale-state trials	[TBD: TBD-HUMAN overtrust]	TBD	Operators notice uncertainty and stale objects

7.8 Low-Visibility and Sensor Dependence

Low-visibility claims must remain out of the main contribution unless sensor-specific evidence is collected. The current paper can define the protocol now, but the claim requires hardware and environmental trials.

Table 11 Low-visibility validation placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
RGB night or glare	Raw video and semantic display	[TBD: TBD-LOWVIS RGB]	TBD	Missed detections explicitly represented
Sensor fallback	Thermal, LiDAR, radar, or equivalent	[TBD: TBD-LOWVIS sensor fallback]	TBD	Sensor limits reported by condition
Uncertainty display	Display variants	[TBD: TBD-LOWVIS uncertainty]	TBD	Operator can identify missing or stale state

8. Limitations and Safety Envelope

Pipeline B can mislead an operator if its uncertainty is hidden. The following limitations are part of the current system definition:

- False negatives are safety-critical because absent objects may simply be undetected.
- Stale actors can look authoritative unless age and confidence are visualized.
- A visually stable synthetic scene can increase overtrust.
- Object metadata, timestamps, and geolocation can still be privacy-sensitive.
- Night, smoke, fog, haze, glare, and occlusion reduce sensor performance unless alternative sensors are validated.
- Cesium, map, terrain, and home-origin errors can become actor-placement errors.
- BVLOS use requires detect-and-avoid, command-and-control reliability, operational-design-domain limits, and regulatory compliance beyond this display.
- VR can introduce simulator sickness, attentional tunneling, fatigue, and display occlusion that must be considered in the human study.

Before any safety-relevant claim, the following thresholds must be defined and measured:

- maximum stale-object age [TBD: TBD-SAFETY max stale age];
- maximum position uncertainty radius [TBD: TBD-SAFETY max uncertainty radius];
- minimum detection confidence for confirmed actor display [TBD: TBD-SAFETY minimum confidence];
- fallback rule when telemetry, detections, or pose become stale [TBD: TBD-SAFETY fallback rule];
- allowed operating envelope by altitude, range, speed, terrain, lighting, and sensor configuration [TBD: TBD-SAFETY allowed operating envelope].

9. VRIH Submission Package Still Required

Before submission, the working draft must be converted to the VRIH author template and accompanied by the journal's required submission materials. The following items can be prepared now:

- title page with all author affiliations and corresponding author details [TBD: author affiliations and corresponding author];
- abstract under 250 words, already satisfied in this draft;
- one to seven keywords, already satisfied in this draft;

- highlights file with three to five concise bullet points [TBD: VRIH highlights file];
- graphical abstract or representative visual, if used [TBD: VR digital-twin screenshot];
- author contribution statement using CRediT roles [TBD: CRediT roles];
- conflict-of-interest statement [TBD: declarations];
- data and code availability statement [TBD: data/code availability];
- ethics statement if human-operator tests are performed [TBD: human-study ethics requirement].

10. Conclusion

Pipeline B is best framed for VRIH as a human-in-the-loop VR digital-twin operator display for dynamic UAV scene reconstruction from in-flight detections. Its current contribution is the implemented integration path from detection messages through Brain state publication into Unreal/Cesium actor reconstruction, together with a concrete validation protocol for measuring whether the approach improves operator awareness under degraded communication conditions.

The manuscript is not yet a completed experimental paper. The work that can be completed immediately is now explicit: VRIH-aligned framing, runtime contract, instrumentation plan, bandwidth benchmark design, partial latency instrumentation, packet-loss testing, and synthetic tracking evaluation. The remaining publication-critical evidence requires hardware: HMD demonstration, source-to-headset latency, calibrated geospatial error, human-operator study, and any low-visibility sensor claims.

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